

24 D Excess weight in one arm over the other = mg
 $= \rho Ahg$
 $= 960 \times 4.0 \times 10^{-4} \times 0.020 \times 9.81 = 0.07534 \text{ N}$

Magnetic force = BI = 0.07534
 $0.07534 = B(60)(0.020)$
 $B = 0.063 \text{ T}$